

# HAOZHE (JIMMY) JIA

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## EDUCATION

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### Boston University

*Master of Science in Computer Science*

Boston, MA

*Sep. 2025 – May 2027*

### Macau University of Science and Technology

*Bachelor of Science in Computer Science*

Taipa, Macao SAR

*Sep. 2021 – Aug. 2025*

## EXPERIENCE

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### Kolachalama Laboratory | BU School of Medicine

Boston, MA

*Graduate Research Assistant | PI: Prof. Vijaya B. Kolachalama*

*Jun. 2026 – Present*

- Developing vision-language models for whole-slide histopathology (WSI), targeting interpretable prediction in Alzheimer's disease neuropathology under NIA-AA ABC staging (Thal, Braak, CERAD).
- Ran proof-of-concept training of vision encoders and Vision Transformer backbones on WSI patch data; designed immunohistochemistry-aware augmentation after establishing that standard photometric augmentation corrupts the DAB stain-intensity signal.
- Benchmarked self-supervised encoders (DINOv2/v3, UNI, Virchow) against caption-supervised pathology encoders (CONCH, PLIP), evaluated by linear probing on held-out splits against a stain-intensity baseline.
- Rebuilt baseline evaluation pipeline components: authored the system prompt for direct question-answering mode, and diagnosed an index misalignment in eval-set preprocessing that was silently dropping rows.

### Bio-Imaging & Informatics Lab | BU School of Medicine

Boston, MA

*Graduate Research Assistant | PI: Dr. Bang-Bon Koo*

*Nov. 2025 – Present*

- Developing an end-to-end deep learning framework (PyTorch, MONAI) for automated perivascular space segmentation from brain MRI, supporting dementia biomarker research across regulated cohorts (ADNI, NACC, OASIS).
- Coordinated the BU-NUS Research Collaboration Agreement securing external validation cohorts from the MICCAI 2024 EPVS Challenge: drafted agreements, negotiated data-access terms, liaised with university governance.
- Designed two-tier evaluation spanning voxel-level metrics (Dice, IoU, HD95) and clinical-agreement analysis (Bland-Altman) against radiologist annotations; containerized pipelines (Docker, Singularity) for reproducible runs on BU SCC.

### BU Spark! | Boston Public Library Digital Commonwealth

Boston, MA

*Machine Learning Engineer | [GitHub](#)*

*Feb. 2026 – May 2026*

- Built a hybrid retrieval-augmented generation system over the Boston Public Library Digital Commonwealth archive (1.35M+ records, 18,601 newspaper issues), combining BGE-M3 dense and sparse embeddings with Neo4j GraphRAG.
- Implemented cited generation grounding every GPT-4o answer in a retrievable archival source, with query logging and metric-computation scripts for retrieval-quality assessment.
- Characterized OCR noise patterns and date-range distribution across a heterogeneous corpus to inform document preparation; drove stack selection (PostgreSQL with pgvector, Neo4j).

### Kingsoft Office Software

*Applied Machine Learning Intern | AI Applied Algorithms Team*

Zhuhai, China

*May 2024 – Jul. 2024*

- Built prompt-engineering and intent-extraction pipelines over internal LLM Gateway APIs, converting noisy user input into structured Excel automation workflows and cutting manual processing from ~1 day to ~3 hours.
- Implemented a multi-label NLP classifier (BERT, TextCNN) across five dialogue types with cross-validation tuning and text-normalization preprocessing.

## SELECTED PROJECTS

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### **MYGO-Centiloid: Cross-Tracer Amyloid PET Quantification** | *PyTorch, MONAI, 3D CNNs* | [GitHub](#) Apr. 2026

- Led a 6-person team to **1st place** at the inaugural BU MedAI Hackathon on the amyloid PET Centiloid regression challenge, achieving best-in-challenge MAE on the ABPET leaderboard.
- Designed a 3D ResNet-18 regressor with TracerNorm conditioning for cross-tracer generalization; ablation identified TracerNorm as essential to cross-tracer robustness.
- Code adopted into the BU Kolachalama Lab (vkola-lab) GitHub and HuggingFace organizations, with formal handoff to lab maintainers.

## HONORS, AWARDS & CERTIFICATIONS

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|-------------------------------------------------------------------------------------------------------|-----------|
| <b>1st Place, BU MedAI Hackathon</b>   BU Faculty of Computing & Data Sciences, Hariri Institute      | Apr. 2026 |
| <b>National Bronze Award</b> , China “Internet+” Innovation & Entrepreneurship Competition (13th ed.) | May 2023  |
| CITI Good Clinical Practice (Biomedical); BUMC HIPAA and Research Data Security                       | 2025      |

## SKILLS

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**Languages & Frameworks:** Python, PyTorch, MONAI, HuggingFace Transformers, vLLM, NumPy, pandas, SQL  
**Machine Learning:** Self-supervised learning (DINO, iBOT), vision transformers, vision-language models, LLM fine-tuning and evaluation, RAG, prompt engineering  
**Tools & Infrastructure:** Linux, Docker, Singularity, HPC job scheduling (SGE, Slurm), Git, AWS, GCP, Neo4j, pgvector  
**Domain:** Medical imaging (MRI, PET, whole-slide pathology), computational pathology, neuroimaging pipelines, HIPAA-compliant and regulated-data workflows